

Thermodynamic approach and Saddle-Dominated Capacity of the LSE Dense Associative Memory

Anonymous submission

Abstract

The Dense Associative Memory with log-sum-exp (LSE) kernel on the N -sphere stores $M = \exp(\alpha N)$ patterns at load α . A Gaussian approximation of the inter-pattern alignments gives a zero-temperature capacity $\alpha_c = 1/2$, independent of the kernel sharpness β , the Gaussian-ensemble threshold of Lucibello and Mézard (2024). Using the exact spherical alignment density, we build a finite-temperature theory of retrieval. The sum over non-target patterns is then set by an interior saddle rather than by the right tail of the Gaussian, which restores the β -dependence: at $T = 0$ the capacity grows from $\sim \beta$ at small β to $\frac{1}{2} \ln \beta$ at large β , is ≈ 0.6226 at $\beta = 1$, and reproduces the exact spherical threshold of Lucibello and Mézard. At $T > 0$ the retrieval basin is bounded by a single-pattern spinodal, beyond which it no longer exists; below it the basin is metastable, with a Kramers escape time exponential in N . We confirm the resulting (α, T) stability diagram by Monte Carlo: a direct sampler stores all M patterns but reaches only $N \sim 20$, while a truncated sampler that keeps only the boundary-setting patterns reaches $N \sim 30$ and agrees with both the direct sampler and the predicted spinodal.

Introduction

Modern dense associative memory (DAM) models begin with Krotov and Hopfield (2016), who generalized the classical Hopfield network (Hopfield 1982) from pairwise to polynomial interactions, raising the storage capacity from linear to polynomial in the number of neurons N while preserving the ability to retrieve previously stored information from noisy or corrupted input data. Taking the polynomial degree to infinity yields an exponential capacity (Demircigil et al. 2017): for binary neurons (state space $\{-1, +1\}^N$), the limiting capacity is $M_c = \exp(\alpha_c N)$ with load $\alpha_c = (\ln 2)/2 \approx 0.35$.

The same exponential scaling carries over to continuous states. Ramsauer et al. (2021) introduced a modern Hopfield network whose state is a continuous vector in $\mathbb{R}^N$ and whose energy is a log-sum-exp (LSE) of the state-pattern overlaps. We take both the stored patterns ξ^μ and the retrieval state $\mathbf{x}$ on the sphere $|\mathbf{x}|^2 = N$, and store $M = \exp(\alpha N)$ patterns at load α . The energy then reads

$$H(\mathbf{x}) = -\beta^{-1} \ln \sum_{\mu=1}^M \exp[-\beta N(1 - \phi^\mu)], \quad (1)$$

where β is a kernel sharpness controlling how strongly the energy distinguishes patterns, and ϕ^μ is the *alignment* of the current state $\mathbf{x}$ with the μ -th pattern ξ^μ ,

$$\phi^\mu(\mathbf{x}) = N^{-1} \mathbf{x} \cdot \xi^\mu. \quad (2)$$

All these models are energy-based. What sets the LSE energy apart from the classical, polynomial, and exponential models is the logarithm: it acts as a smooth maximum, keeping the energy extensive in N rather than exponentially large, while its gradient is a softmax over patterns, so the retrieval update replaces the state by a softmax-weighted average of the stored patterns, exactly like the attention mechanism of transformers. Model (1) is the Gaussian-kernel member of a family of kernel-based energies. Replacing the Gaussian by the compactly supported Epanechnikov kernel gives the log-sum-ReLU (LSR) model (Hoover et al. 2025).

A natural question is how retrieval survives thermal noise. At a finite temperature T of the thermal bath (distinct from the kernel sharpness β), the boundary $\alpha_c(\beta, T)$ in the load-temperature plane separates cues that relax back to their target from cues that escape to spurious states. Petrova, Polyachenko, and State (2026) derived this boundary for the LSE and LSR kernels under a Gaussian approximation of the inter-pattern alignment density, obtaining for LSE a β -independent zero-temperature load $\alpha_c = 1/2$, and identifying the Gaussian approximation as the likely source of error. Direct Monte Carlo (Petrova 2026) then showed deviations from the Gaussian prediction, with the retrieval region extending to larger α as $T \rightarrow 0$.

Here we revisit the LSE capacity on the N -sphere using the exact spherical alignment density, both analytically and by Monte Carlo, across the full (α, T) plane. The sum over non-target patterns is governed by an interior saddle of the partition-sum integrand rather than by the right edge of the alignment distribution, which restores the β -dependence missing from the Gaussian estimate. At $T = 0$ the resulting capacity reproduces the exact spherical threshold of Lucibello and Mézard (2024) and lies above the rigorous storage lower bound of Ramsauer et al. (2021). At $T > 0$ the retrieval well is bounded by a spinodal, the limit of metastability beyond which the well disappears; below it the well persists metastably, and a trapped cue escapes only over a barrier, with a lifetime exponential in N , a Kramers escape (Kramers 1940; Hänggi, Talkner, and Borkovec 1990). This resolves the low- T deviations seen in simulations.

The paper is organized as follows. The Theory section builds the finite-temperature retrieval theory from the exact spherical alignment density: the Gaussian baseline and why it fails, the interior saddle that fixes the noise floor, the zero-temperature capacity, and the finite-temperature spinodal. The Monte Carlo schemes section describes the direct and truncated samplers. The Results section presents the stability diagram in the (α, T) plane at $\beta = 1$, compares it against the Gaussian and spinodal predictions, and examines the finite- N behavior. The Discussion section collects the implications and the relation to the literature.

Loss of the retrieval in the large- N limit

Let the retrieval pattern ξ^1 and a spurious pattern ξ be drawn independently and uniformly from the sphere of radius $\sqrt{N}$ in $\mathbb{R}^N$. The alignment

$$\varphi(\xi) \equiv N^{-1} \xi^1 \cdot \xi \quad (3)$$

is the cosine of the angle between them. Because the alignment depends only on the relative orientation of the two vectors, we may fix the first pattern along the coordinate axis e_1 and treat the spurious pattern as a unit vector $\mathbf{u}$ uniformly distributed on S^{N-1} . The alignment is then simply its first coordinate, $\varphi = u_1$, and we seek the density of u_1 induced by the uniform measure on the sphere. The probability that u_1 lies in $[\varphi, \varphi + d\varphi]$ equals the area of the corresponding spherical band divided by the total area of S^{N-1} . Decomposing $\mathbf{u} = (u_1, \mathbf{u}_\perp)$, the constraint $|\mathbf{u}| = 1$ fixes $|\mathbf{u}_\perp| = \sqrt{1 - \varphi^2}$, so the points at height φ form a sphere S^{N-2} of radius $\sqrt{1 - \varphi^2}$, with area proportional to $(1 - \varphi^2)^{(N-2)/2}$. The band thickness is $d\varphi/\sqrt{1 - \varphi^2}$, since the surface is inclined with respect to the axis. Normalizing the band area $\propto (1 - \varphi^2)^{(N-3)/2} d\varphi$ gives the density

$$h(\varphi) = C_N (1 - \varphi^2)^{(N-3)/2} \quad (4)$$

with the normalization constant

$$C_N = \frac{\Gamma(N/2)}{\sqrt{\pi} \Gamma((N-1)/2)}. \quad (5)$$

The density is symmetric about $\varphi = 0$, with mean zero and variance $1/N$. As $N \rightarrow \infty$ it concentrates near the equator and approaches a Gaussian,

$$h(\varphi) \longrightarrow \sqrt{\frac{N}{2\pi}} e^{-N\varphi^2/2}, \quad (6)$$

so distinct high-dimensional patterns are nearly orthogonal: typical alignments are of order $N^{-1/2}$. The exact law is retained where the rare large-alignment tail matters, the Gaussian form being adequate only near the center of the distribution. The density (4) and its Gaussian limit are the spherical and Gaussian rate functions of Lucibello and Mézard (2024).

Ramsauer et al. (2021) tie retrieval to the separation of a pattern from its nearest neighbour, $N(1 - \varphi_{\max})$, where $\varphi_{\max}$ is the highest alignment in the set of $M = e^{\alpha N}$ patterns (their Theorem 4): the retrieval pattern is lost when a spurious pattern reaches it, $\varphi_{\max} \rightarrow 1$. The load at which this happens is fixed by the alignment law $h(\varphi)$ near $\varphi = 1$.

Under the Gaussian approximation h keeps weight at $\varphi = 1$, and $\varphi_{\max}$ reaches 1 at the β -independent load $\alpha_c^G = 1/2$. Under the exact law (4) h vanishes at $\varphi = 1$, since two distinct patterns cannot align perfectly on the sphere, so $\varphi_{\max} = \sqrt{1 - e^{-2\alpha}} < 1$ for every α and the capacity is infinite. Both predictions are unsatisfactory: $\alpha_c^G = 1/2$ carries no dependence on the kernel sharpness β , and the exact nearest-pattern estimate gives no finite capacity.

We now analyse retrieval through the thermodynamic approach using the free energy density profile. The latter includes the internal energy density $u = H/N$ and the entropy term. The softmax sum $S(\mathbf{x})$ from (1) includes the retrieval basin term, and the sum over spurious patterns,

$$S(\mathbf{x}) \equiv \exp[\beta N(\varphi(\mathbf{x}) - 1)] + S_{\text{spur}}(\mathbf{x}), \quad (7)$$

$$S_{\text{spur}}(\mathbf{x}) \equiv \sum_{\mu=2}^M \exp[\beta N(\varphi^\mu(\mathbf{x}) - 1)]. \quad (8)$$

The spurious patterns are i.i.d. uniform on the sphere and independent of $\mathbf{x}$, so for any $\mathbf{x}$ with $|\mathbf{x}|^2 = N$ the projections $\varphi^\mu(\mathbf{x}) = \mathbf{x} \cdot \xi^\mu/N$ have marginal density (4). The spurious sum is therefore distributed identically at every $\mathbf{x}$ on the sphere; in particular at $\mathbf{x} = \xi^1$, where the projections become the inter-pattern alignments $\varphi_{1\mu}$. With $M = e^{\alpha N}$, the sum concentrates around its expectation, and

$$S_{\text{spur}} \approx M \int_{-1}^1 d\varphi h(\varphi) \exp[\beta N(\varphi - 1)]. \quad (9)$$

The large- N thermodynamic limit is obtained by a saddle-point evaluation of the integral (9). The saddle φ^* is the solution of the equation $\partial g(\varphi, \beta)/\partial \varphi = 0$, where

$$g(\varphi, \beta) = \ln(1 - \varphi^2)/2 + \beta \varphi. \quad (10)$$

This gives the saddle point at

$$\varphi^*(\beta) = \frac{\sqrt{1 + 4\beta^2} - 1}{2\beta}, \quad (11)$$

and the saddle value

$$g_{\max}(\beta) = g(\varphi^*, \beta) = \frac{1}{2} \ln(1 - \varphi^{*2}) + \beta \varphi^*. \quad (12)$$

The spurious sum without the prefactor is then

$$S_{\text{spur}} \sim \exp[N(\alpha + g_{\max}(\beta) - \beta)]. \quad (13)$$

Appendix A derives the prefactor, which turns out to be a mildly dependent function of $\varphi^*(\beta)$ only; for $\beta = 1$ the prefactor is 1.082. In the thermodynamic limit, however, only the leading term, proportional to N in the exponent survives, so the prefactor is unneeded on this stage. The internal energy density

$$u = \frac{H}{N} = 1 - \varphi \quad (14)$$

when the retrieval term dominates, and

$$u = 1 - \frac{1}{\beta} [\alpha + g_{\max}(\beta)] \quad (15)$$

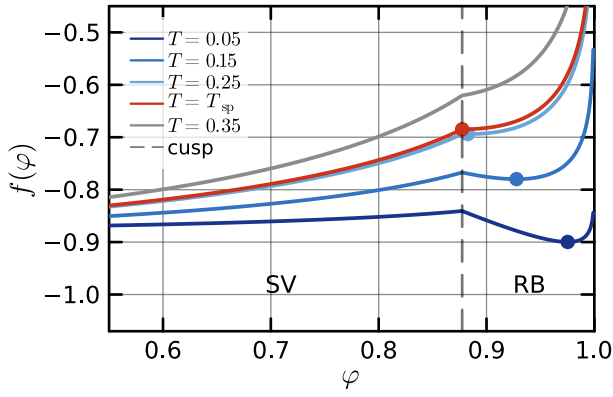


Figure 1: Leading order- N free energy density $f(\varphi)$ at $\alpha = 0.5$ and $\beta = 1$ for different values of T , showing the retrieval basin (RB), the saddle valley (SV) formed by spurious modes predominantly near the saddle point, and the cusp at $\varphi \approx 0.877$ (grey dashes). The color dots show the RB minima; they disappear at the spinodal – critical temperature $T_{\text{sp}}(\alpha, \beta) \approx 0.262$. The limiting capacity is achieved when the cusp reaches $\varphi = 1$.

when the spurious term dominates. The two terms form a cusp where the two terms switch dominance, at $\varphi_c(\alpha, \beta) = (\alpha + g_{\text{max}}(\beta))/\beta$.

Combining the internal energy with the entropy density term (Petrova, Polyachenko, and State 2026)

$$s = \frac{1}{2} \ln(1 - \varphi^2), \quad (16)$$

one obtains the free energy density $f = u(\varphi) - Ts(\varphi)$, which is plotted in Fig. 1 for $\alpha = 0.5$, $\beta = 1$ and different values of T . The overall profile has 2 minima: one at $\varphi \approx 0$ due to the spurious modes, and one at $\varphi \approx 1$ due to the retrieval term. Which one is global, which one is local depends on the temperature threshold $T_{\text{sd}}(\alpha, \beta)$, however for the retrieval it is not important whether the retrieval minimum is global or local, but whether it exists at all, which is the spinodal – the point where the retrieval minimum disappears.

The retrieval minimum is at

$$\varphi_{\text{eq}} = \frac{1}{2} \left(\sqrt{T^2 + 4} - T \right), \quad (17)$$

so the condition for the spinodal is $\varphi_{\text{eq}} = \varphi_c(\alpha, \beta)$, yielding

$$T_{\text{sp}}(\alpha, \beta) = \frac{\beta^2 - (\alpha + g_{\text{max}}(\beta))^2}{\beta(\alpha + g_{\text{max}}(\beta))}. \quad (18)$$

For $T < T_{\text{sp}}(\alpha, \beta)$, the local and global retrieval minimum is metastable: the system can occasionally leave the basin and wander into the saddle basin through the Kramers process. To do so, the system must overcome the Kramers barrier $\Delta f/T$. In the vanishing T , the retrieval is retained unless the retrieval minimum (local or global) is present.

Ramsauer et al. (2021) replace the spurious sum S_{spur} by its largest term times the count,

$$S_{\text{spur}} \leq M e^{-\beta N(1 - \varphi_{\text{max}})} = e^{N(\alpha - \beta(1 - \varphi_{\text{max}}))}, \quad (19)$$

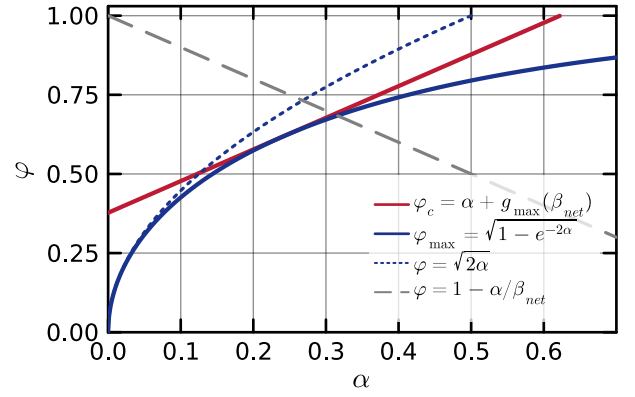


Figure 2: Zero-temperature capacity $\alpha_c^{\text{sd}}(\beta, 0)$ of the LSE DAM as a function of the kernel sharpness β (red), together with Ramsauer's β -independent bound $\alpha_c^{\text{G}} = 1/2$ (blue dashed).

which exceeds the retrieval term once $\varphi_{\text{max}} > 1 - \alpha/\beta$. The highest alignment actually present is $\varphi_{\text{max}} = \sqrt{1 - e^{-2\alpha}}$, so Ramsauer et al.'s capacity is the load where

$$\sqrt{1 - e^{-2\alpha}} = 1 - \frac{\alpha}{\beta}, \quad (20)$$

finite and β -dependent.

Fig. 2 summarises the different approaches to the capacity. The red curve is the zero-temperature capacity obtained from the saddle-point evaluation of the spurious sum which grows with β and reproduces the exact spherical threshold of Lucibello and Mézard (2024) at large β .

The solid/dotted blue lines are the capacities obtained from the highest alignment condition, for the exact spherical law and the Gaussian approximation respectively (incorrect). The Ramsauer et al. capacity is obtained from the intersection of the solid blue line and the grey dashed lines and β -dependent, but underestimates the capacity at all β .

An interesting point is that the red line is always tangent to the blue line. The touching point depends on β and sweeps the whole blue curve as β varies. This is a consequence of the Legendre-conjugacy relation between the saddle value $g_{\text{max}}(\beta)$ and the spherical extreme $\varphi_{\text{max}}(\alpha)$, which is explained in Appendix B. The direct consequence of this is that there is no spurious patterns inside the retrieval basin determined by the position of the cusp (red curve for a given α), while the maximum alignment $\varphi_{\text{max}}(\alpha)$ is given by the solid blue line. This corresponds to (Lucibello and Mézard 2024) claim that the retrieval loss is always *unconditional* (i.e. due to a collective effect of the spurious patterns which we estimated by the saddle-point analysis), opposite to *conditional* loss due to a single spurious pattern reaching the retrieval pattern, which is the one estimated by the extreme-value analysis.

Monte Carlo schemes

Results

Discussion

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Appendix A: Derivation of the prefactor in the spurious sum

Density (4) allows the exact solution in terms of the modified Bessel function of the first kind. It can be obtained by changing variable to $\phi = \cos \theta$ and using the integral representation of $I_\nu(z)$ (Gradshteyn and Ryzhik 2015, 8.431.3), applied with $\nu = N/2 - 1$, $z = \beta N$, gives for $N > 1$

$$S_{\text{spur}} = M \Gamma\left(\frac{N}{2}\right) \left(\frac{2}{\beta N}\right)^{N/2-1} e^{-\beta N} I_{N/2-1}(\beta N). \quad (21)$$

This closed form is the disorder expectation of sum (8). For each particular disorder realisation, S_{spur} concentrates on it with relative fluctuations of order $M^{-1/2} = e^{-\alpha N/2}$, exponentially small in N .

The Debye uniform asymptotic for the modified Bessel function allows to obtain also the prefactor (Olver et al. 2010, 10.41.3). With the lowest $k = 0$ term in the expansion series,

$$I_\nu(\nu z') \sim \frac{e^{\nu \eta(z')}}{(2\pi\nu)^{1/2}(1+z'^2)^{1/4}}, \quad (22)$$

where $z' = 2\beta N/(N-2)$ and

$$\eta(z') = \sqrt{1+z'^2} + \ln \frac{z'}{1+\sqrt{1+z'^2}}. \quad (23)$$

This function should be expanded up to $\mathcal{O}(N^{-1})$ because it is multiplied to the large parameter $\nu = N/2 - 1$ in the exponent:

$$\eta(z') = \eta(2\beta) + \eta'(2\beta) \frac{4\beta}{N} + \mathcal{O}(N^{-2}), \quad (24)$$

where

$$\eta(2\beta) = 1 + 2\beta\phi^* + \ln \phi^*, \quad (25)$$

$$\eta'(2\beta) = \phi^* + \frac{1}{2\beta}. \quad (26)$$

Then

$$\nu \eta(z') = \frac{N}{2} \eta(2\beta) - \ln \phi^* + \mathcal{O}(1/N). \quad (27)$$

So, the $\mathcal{O}(1)$ part of $\nu \eta(z')$ collapses to $-\ln \phi^*$. Using Stirling formulae

$$\Gamma(N/2) \sim \sqrt{4\pi/N} (N/2)^{N/2} e^{-N/2} \quad (28)$$

and collecting all terms in (21) to order $\mathcal{O}(1)$ in the exponent, and retaining the leading order $\mathcal{O}(N)$ in the prefactor, one obtains the same leading-order exponent as in (13), and the final expression that extends (13) with the prefactor, is hence

$$S_{\text{spur}} \approx \frac{e^{N[\alpha + g_{\max}(\beta) - \beta]}}{\sqrt{1 - \phi^{*4}}} \quad (29)$$

with $g_{\max}(\beta)$ given by (12). At $\beta = 1$, $\phi^* \approx 0.618$ and the prefactor equals ≈ 1.082 .

Appendix B: Legendre-conjugacy relation between the saddle value and the spherical extreme

The leading-order cusp $\varphi_c(\alpha, \beta) = (\alpha + g_{\max}(\beta))/\beta$ is, at fixed β , a line in α with slope $1/\beta$ and intercept $g_{\max}(\beta)/\beta$. As β varies, the spherical extreme $\varphi_{\max}(\alpha) = \sqrt{1 - e^{-2\alpha}}$ is the lower envelope of the resulting one-parameter family of lines, and each line is tangent to the curve at the corresponding saddle point $\varphi^*(\beta)$.

The envelope condition $\partial_\beta \varphi_c|_\alpha = 0$ reads $\beta g'_{\max}(\beta) - g_{\max}(\beta) = \alpha$. From $g_{\max}(\beta) = \frac{1}{2} \ln(1 - \varphi^{*2}) + \beta \varphi^*$ and the saddle equation $\beta \varphi^{*2} + \varphi^* - \beta = 0$ one obtains $g'_{\max}(\beta) = \varphi^*$, so the tangency abscissa is

$$\alpha_0(\beta) = \beta \varphi^* - g_{\max}(\beta) = -\frac{1}{2} \ln(1 - \varphi^{*2}). \quad (30)$$

At $\alpha = \alpha_0$ the line takes value $\varphi^*(\beta)$. Squaring (30) yields $\varphi^{*2} = 1 - e^{-2\alpha_0}$, so the point (α_0, φ^*) lies on the spherical-extreme curve. As β sweeps $(0, \infty)$, φ^* sweeps $(0, 1)$ and the tangency sweeps the full curve from the origin to the right edge. The curve is concave (a direct computation gives $\varphi''_{\max} < 0$), so each cusp line lies above it apart from tangency, and $\varphi_{\max}(\alpha) = \inf_{\beta>0} \varphi_c(\alpha, \beta)$. The functions $g_{\max}(\beta)$ and $-\frac{1}{2} \ln(1 - \varphi^2)$ are Legendre conjugates.